```
import random

# 20 tasks execution times
tasks = [10, 5, 8, 12, 7, 6, 15, 4, 9, 11,
         3, 14, 6, 8, 5, 13, 7, 4, 10, 6]

VM = 4
particles = 10
iterations = 50

# Create random task allocation
swarm = [[random.randint(0, VM-1) for _ in tasks]
         for _ in range(particles)]

pbest = [p[:] for p in swarm]
pbest_value = [float('inf')] * particles

gbest = None
gbest_value = float('inf')

for _ in range(iterations):

    for i in range(particles):

        # Calculate VM loads
        load = [0] * VM

        for task, vm in zip(tasks, swarm[i]):
            load[vm] += task

        makespan = max(load)

        # Update personal best
        if makespan < pbest_value[i]:
            pbest_value[i] = makespan
            pbest[i] = swarm[i][:]

        # Update global best
        if makespan < gbest_value:
            gbest_value = makespan
```

```
            gbest = swarm[i][:]

        # Randomly move tasks towards best solution
        for j in range(len(tasks)):
            if random.random() < 0.5:
                swarm[i][j] = pbest[i][j]

            if random.random() < 0.5:
                swarm[i][j] = gbest[j]

            # Random exploration
            if random.random() < 0.1:
                swarm[i][j] = random.randint(0, VM-1)

# Final VM loads
load = [0] * VM

for task, vm in zip(tasks, gbest):
    load[vm] += task

print("Best Task Allocation:", gbest)
print("VM Loads:", load)
print("Minimum Makespan:", gbest_value)
```

Output:

```
Best Task Allocation: [3, 3, 3, 2, 3, 2, 0, 1, 1, 3, 0, 0, 2, 1, 0, 1, 1, 0, 2, 2]
VM Loads: [41, 41, 40, 41]
Minimum Makespan: 41
```